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State Space Analysis

GATE ECE Control Systems · Previous Year Questions with Answer Key

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Marks 1 5 questions

Q1. GATE ECE 2017 Set 2

Consider the state space realization

$$\begin{bmatrix} \dot{x}_1 & (t) \\ \dot{x}_2 & (t) \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & -9 \end{bmatrix} \begin{bmatrix} x_1 & (t) \\ x_2 & (t) \end{bmatrix} + \begin{bmatrix} 0 \\ 45 \end{bmatrix} u\,(t),$$

with the initial condition $\begin{bmatrix} x_1 & (0) \\ x_2 & (0) \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$;
Where u(t) denotes the unit step function.
The value of $\underset{t\to\infty}{Lt} \left| \sqrt{{x_1}^2\,(t) + {x_2}^2\,(t)} \right|$ is _____

Open solution

Q2. GATE ECE 2009

Consider the system $\frac{dx}{dt} = Ax + Bu$ with $\mathrm{A} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$ and $\mathrm{B} = \begin{bmatrix} p \\ q \end{bmatrix}$,
where p and q are arbitrary real numbers. Which of the following statesments about the controllability of the system is true?

(A) The system is completely state controllable for any nonzero values of p and q.

(B) Only p =0 and q=0 result in controllability.

(C) The system is uncontrollable for all values of p and q.

(D) We cannot conclude about controllability from the given data.

Open solution

Q3. GATE ECE 2005

A linear system is equivalently represented by two sets of state equations. $\dot{x} = \mathrm{A}X + BU$ and $\dot{W} = CW + DU$. The eigen values of the representations are also computed as $[\lambda]$ and $[\mu]$. Which one of the following statements is true?

(A) $[\lambda] = [\mu]\ and\ X = W$

(B) $[\lambda] = [\mu]\ and\ X \neq W$

(C) $[\lambda] \neq [\mu]\ and\ X = W$

(D) $[\lambda] \neq [\mu]\ and\ X \neq W$

Open solution

Q4. GATE ECE 2002

The transfer function Y(s)/U(s) of a system described by the state equations
$\dot{x}$(t) = -2x(t)+2u(t)
y(t) = 0.5x(t) is

(A) $\frac{0.5}{(s-2)}$

(B) $\frac{1}{(s-2)}$

(C) $\frac{0.5}{s+2}$

(D) $\frac{1}{s+2}$

Open solution

Q5. GATE ECE 1999

The system mode described by the state equations

$$X = \begin{pmatrix} 0 & 1 \\ 2 & -3 \end{pmatrix} x + \begin{pmatrix} 0 \\ 1 \end{pmatrix} u, y = [1 \quad 1]x$$

(A) controllable and observable

(B) controllable, but not observable

(C) observable, but not controllable

(D) neither controllable nor observable

Open solution

Marks 2 33 questions

Q6. GATE ECE 2026

The state and output equations for a control system are:

$$\dot{x} = \begin{bmatrix} -4 & -1.5 \\ 4 & 0 \end{bmatrix} x + \begin{bmatrix} 2 \\ 0 \end{bmatrix} u$$
$$y = [1.5 \quad 0.625]x$$

Which of the following expressions correctly represents the transfer function $\frac{Y(s)}{U(s)}$ of the system with zero initial conditions?

(A) $\frac{3s}{s^2+4s-6}$

(B) $\frac{3s+5}{s^2+4s-6}$

(C) $\frac{3s+5}{s^2+4s+6}$

(D) $\frac{3s}{s^2+4s+6}$

Open solution

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Q7. GATE ECE 2025

Open solution

Consider a system where $x_1(t), x_2(t)$, and $x_3(t)$ are three internal state signals and $u(t)$ is the input signal. The differential equations governing the system are given by

$$\frac{d}{dt}\begin{bmatrix} x_1(t) \\ x_2(t) \\ x_3(t) \end{bmatrix} = \begin{bmatrix} 2 & 0 & 0 \\ 0 & -2 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1(t) \\ x_2(t) \\ x_3(t) \end{bmatrix} + \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} u(t)$$

Which of the following statements is/are TRUE?

(A) The signals $x_1(t), x_2(t)$, and $x_3(t)$ are bounded for all bounded inputs.

(B) There exists a bounded input such that at least one of the signals $x_1(t), x_2(t)$, and $x_3(t)$ is unbounded.

(C) There exists a bounded input such that the signals $x_1(t), x_2(t)$ and $x_3(t)$ are unbounded.

(D) The signals $x_1(t), x_2(t)$ and $x_3(t)$ are unbounded for all bounded inputs.

Q8. GATE ECE 2024

Open solution

Consider a system S represented in state space as

$$\frac{dx}{dt} = \begin{bmatrix} 0 & -2 \\ 1 & -3 \end{bmatrix} x + \begin{bmatrix} 1 \\ 0 \end{bmatrix} r, \quad y = [2 \quad -5]x.$$

Which of the state space representations given below has/have the same transfer function as that of S?

(A) $\frac{dx}{dt} = \begin{bmatrix} 0 & 1 \\ -2 & -3 \end{bmatrix} x + \begin{bmatrix} 0 \\ 1 \end{bmatrix} r, \quad y = [1 \quad 2]x.$

(B) $\frac{dx}{dt} = \begin{bmatrix} 0 & 1 \\ -2 & -3 \end{bmatrix} x + \begin{bmatrix} 1 \\ 0 \end{bmatrix} r, \quad y = [0 \quad 2]x.$

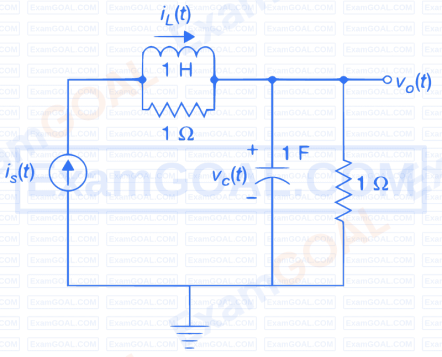
(C) $\frac{dx}{dt} = \begin{bmatrix} -1 & 0 \\ 0 & -2 \end{bmatrix} x + \begin{bmatrix} -1 \\ 3 \end{bmatrix} r, \quad y = [1 \quad 1]x.$

(D) $\frac{dx}{dt} = \begin{bmatrix} -1 & 0 \\ 0 & -2 \end{bmatrix} x + \begin{bmatrix} 1 \\ 1 \end{bmatrix} r, \quad y = [1 \quad 2]x.$

Q9. GATE ECE 2021

Open solution

The electrical system shown in the figure converts input source current $i_s(t)$ to output voltage $\theta_O(t)$.



Current $i_L(t)$ in the inductor and voltage $\vartheta_C(t)$ across the capacitor ate taken as the state variables, both assumed to be initially equal to Zero, i.e., $i_L(0) = 0$ and $\vartheta_c(0) = 0$. The system is

(A) neither state controllable nor observable

(B) completely state controllable but not observable

(C) completely observable but not state controllable

(D) completely state controllable as well as completely observable

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Q10. GATE ECE 2018

Open solution

The state equation and the output equation of a control system are given below:

$$\dot{x} = \begin{bmatrix} -4 & -1.5 \\ 4 & 0 \end{bmatrix} x + \begin{bmatrix} 2 \\ 0 \end{bmatrix} u,$$

$$y = \begin{bmatrix} 1.5 & 0.625 \end{bmatrix} x.$$

The transfer function representation of the system is

(A) $\frac{3s+5}{s^2+4s+6}$
(B) $\frac{3s-1.875}{s^2+4s+6}$
(C) $\frac{4s+1.5}{s^2+4s+6}$
(D) $\frac{6s+5}{s^2+4s+6}$

Q11. GATE ECE 2017 Set 2

Open solution

A second order LTI system is described by the following state equation.

$$\frac{d}{dt} x_1(t) - x_2(t) = 0$$
$$\frac{d}{dt} x_2(t) + 2x_1(t) + 3x_2(t) = r(t)$$

When $x_1(t)$ and $x_2(t)$ are the two state variables and r(t) denotes the input. The output $c(t)=X_1(t)$. The systyem is

(A) undamped (oscillatory)
(B) under damped
(C) critically damped
(D) over damped

Q12. GATE ECE 2016 Set 3

Open solution

A second-order linear time-invariant system is described by the following state equations

$$\frac{d}{dt} x_1(t) + 2x_1(t) = 3u(t)$$
$$\frac{d}{dt} x_2(t) + x_2(t) = u(t)$$

where $x_1(t)$ and $x_2(t)$ are the state variables and u(t) is the input. If the output $y(t) = x_1(t)$, then the system is

(A) controllable but not observable
(B) observable but not controllable
(C) both controllable and observable
(D) neither controllable nor observable

Q13. GATE ECE 2015 Set 3

Open solution

A network is described by the state model as

$$\dot{x}_1 = 2x_1 - x_2 + 3u,$$
$$\dot{x}_2 = -4x_2 - u,$$
$$y = 3x_1 - 2x_2$$

the transfer function H(s) $\left[= \frac{Y(s)}{U(s)}\right] is$

(A) $\frac{11s+35}{(s-2)(s+4)}$
(B) $\frac{11s-35}{(s-2)(s+4)}$
(C) $\frac{11s+38}{(s-2)(s+4)}$
(D) $\frac{11s-38}{(s-2)(s+4)}$

Q14. GATE ECE 2015 Set 2

Open solution

The state variable representation of a system is given as

$$\dot{x} = \begin{bmatrix} 0 & 1 \\ 0 & -1 \end{bmatrix} x; x(0) = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$
$$y = \begin{bmatrix} 0 & 1 \end{bmatrix} x$$

The response y(t) is

(A) sin(t)
(B) $1\text{-}e^t$
(C) 1-cos(t)
(D) 0

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Q15. GATE ECE 2014 Set 1

Open solution

Consider the state space model of a system, as given below

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \end{bmatrix} = \begin{bmatrix} -1 & 1 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & -2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} + \begin{bmatrix} 0 \\ 4 \\ 0 \end{bmatrix} u;$$

$$y = \begin{bmatrix} 1 & 1 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}.$$

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The system is

(A) controllable and observable.

(B) uncontrollable and observable.

(C) uncontrollable and unobservable.

(D) controllable and unobservable.

Q16. GATE ECE 2014 Set 4

Open solution

The state transition matrix $\phi(t)$ of a system

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} is$$

(A) $\begin{bmatrix} t & 1 \\ 1 & 0 \end{bmatrix}$

(B) $\begin{bmatrix} 1 & 0 \\ t & 1 \end{bmatrix}$

(C) $\begin{bmatrix} 0 & 1 \\ 1 & t \end{bmatrix}$

(D) $\begin{bmatrix} 1 & t \\ 0 & 1 \end{bmatrix}$

Q17. GATE ECE 2014 Set 2

Open solution

An unforced linear time invariant (LTI) system is represented by

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} -1 & 0 \\ 0 & -2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}.$$

If the initial conditions are $x_1(0)= 1$ and $x_2(0)=-1$, the solution of the state equation is

(A) $x_1(t) = -1, x_2(t) = 2$

(B) $x_1(t) = -e^{-t}, x_2(t) = 2e^{-t}$

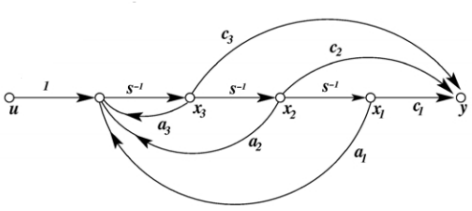
(C) $x_1(t) = e^{-t}, x_2(t) = -e^{-2t}$

(D) $x_1(t) = e^{-t}, x_2(t) = -2e^{-t}$

Q18. GATE ECE 2014 Set 2

Open solution

Consider the state space system expressed by the signal flow diagram shown in the figure.



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The corresponding system is

(A) always controllable

(B) always observable

(C) always stable

(D) always unstable

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Q19. GATE ECE 2014 Set 3

Open solution

The state equation of a second-order linear system is given by $\dot{x}(t) = Ax(t), x(0) = x_0$.

For $x_0 = \begin{bmatrix} 1 \\ -1 \end{bmatrix}, x(t) = \begin{bmatrix} e^{-t} \\ -e^{-t} \end{bmatrix}$ and for $x_0 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}, x(t) = \begin{bmatrix} e^{-t} & -e^{-2t} \\ -e^{-t} & +2e^{-2t} \end{bmatrix}$ when $x_0 = \begin{bmatrix} 3 \\ 5 \end{bmatrix}, x(t)$ is

(A) $\begin{bmatrix} -8e^{-t} & +11e^{-2t} \\ 8e^{-t} & -22e^{-2t} \end{bmatrix}$

(B) $\begin{bmatrix} 11e^{-t} & -8e^{-2t} \\ -11e^{-t} & +16e^{-2t} \end{bmatrix}$

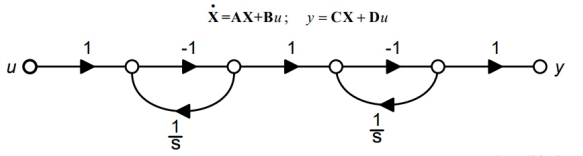
(C) $\begin{bmatrix} 3e^{-t} & -5e^{-2t} \\ -3e^{-t} & +10e^{-2t} \end{bmatrix}$

(D) $\begin{bmatrix} 5e^{-t} & -3e^{-2t} \\ -5e^{-t} & +6e^{-2t} \end{bmatrix}$

Q20. GATE ECE 2013

Open solution

The state diagram of a system is shown below. A system is shown below. A system is described by the state variable equations



The state-variable equations of the system shown in the figure above are

(A) $\dot{X} = \begin{bmatrix} -1 & 0 \\ 1 & -1 \end{bmatrix} X + \begin{bmatrix} -1 \\ 1 \end{bmatrix} u$

$y = [1 \quad -1]X + u$

(B) $\dot{X} = \begin{bmatrix} -1 & 0 \\ -1 & -1 \end{bmatrix} X + \begin{bmatrix} -1 \\ 1 \end{bmatrix} u$

$y = [-1 \quad -1]X + u$

(C) $\dot{X} = \begin{bmatrix} -1 & 0 \\ -1 & -1 \end{bmatrix} X + \begin{bmatrix} -1 \\ 1 \end{bmatrix} u$

$y = [-1 \quad -1]X - u$

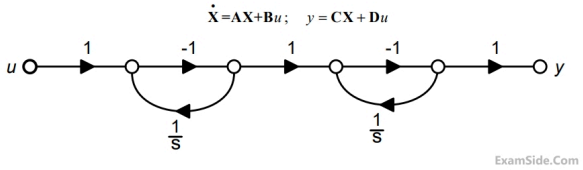
(D) $\dot{X} = \begin{bmatrix} -1 & -1 \\ 0 & -1 \end{bmatrix} X + \begin{bmatrix} -1 \\ 1 \end{bmatrix} u$

$y = [1 \quad -1]X - u$

Q21. GATE ECE 2013

Open solution

The state diagram of a system is shown below. A system is described by the state variable equations



The state transition matrix e^{At} of the system shown in the figure above is

(A) $\begin{bmatrix} e^{-t} & 0 \\ te^{-t} & e^{-t} \end{bmatrix}$

(B) $\begin{bmatrix} e^{-t} & 0 \\ -te^{-t} & e^{-t} \end{bmatrix}$

(C) $\begin{bmatrix} e^{-t} & 0 \\ e^{-t} & e^{-t} \end{bmatrix}$

(D) $\begin{bmatrix} e^{-t} & -te^{-t} \\ 0 & e^{-t} \end{bmatrix}$

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Q22. GATE ECE 2012

Open solution

The state variable description of an LTI system is given by

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \end{bmatrix} = \begin{bmatrix} 0 & a_1 & 0 \\ 0 & 0 & a_2 \\ a_3 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} u,$$

$$y = \begin{pmatrix} 1 & 0 & 0 \end{pmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

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where y is the output and u is input. The system is controllable for

(A) $a_1 \neq 0, a_2 = 0, a_3 \neq 0$

(B) $a_1 = 0, a_2 \neq 0, a_3 \neq 0$

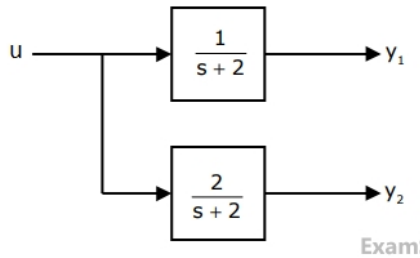
(C) $a_1 = 0, a_2 \neq 0, a_3 = 0$

(D) $a_1 \neq 0, a_2 \neq 0, a_3 = 0$

Q23. GATE ECE 2011

Open solution

The block diagram of a system with one input u and two outputs y_1 and y_2 is given below



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A state space model of the above system in terms of the state vector $\underline{x}$ and the output vector $\underline{y} = [y_1 \quad y_2]^T$ is

(A) $\dot{\underline{x}} = [2]\underline{x} + [1]u; \underline{y} = [1 \quad 2]x$

(B) $\dot{\underline{x}} = [-2]\underline{x} + [1]u; \underline{y} = \begin{bmatrix} 1 \\ 2 \end{bmatrix} x$

(C) $\dot{\underline{x}} = \begin{bmatrix} -2 & 0 \\ 0 & -2 \end{bmatrix} \underline{x} + \begin{bmatrix} 1 \\ 1 \end{bmatrix} u; \underline{y} = [1 \quad 2]x$

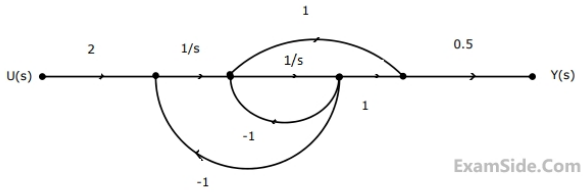
(D) $\dot{\underline{x}} = \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix} \underline{x} + \begin{bmatrix} 1 \\ 1 \end{bmatrix} u; \underline{y} = \begin{bmatrix} 1 \\ 2 \end{bmatrix} x$

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Q24. GATE ECE 2010

Open solution

The signal flow graph of a system is shown below.



The state variable representation of the system can be

(A)
$$\dot{x} = \begin{bmatrix} 1 & 1 \\ -1 & 0 \end{bmatrix} x + \begin{bmatrix} 0 \\ 2 \end{bmatrix} u$$
$$y = [0 \quad 0.5]x$$

(B) $\dot{x} = \begin{bmatrix} -1 & 1 \\ -1 & 0 \end{bmatrix} x + \begin{bmatrix} 0 \\ 2 \end{bmatrix} u$
$$y = [0 \quad 0.5]x$$

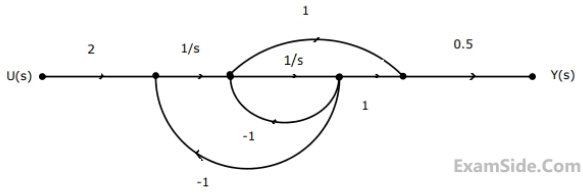
(C) $\dot{x} = \begin{bmatrix} 1 & 1 \\ -1 & 0 \end{bmatrix} x + \begin{bmatrix} 0 \\ 2 \end{bmatrix} u$
$$y = [0.5 \quad 0.5]x$$

(D) $\dot{x} = \begin{bmatrix} -1 & 1 \\ -1 & 0 \end{bmatrix} x + \begin{bmatrix} 0 \\ 2 \end{bmatrix} u$
$$y = [0.5 \quad 0.5]x$$

Q25. GATE ECE 2010

Open solution

The signal flow graph of a system is shown below.



The transfer function of the system is

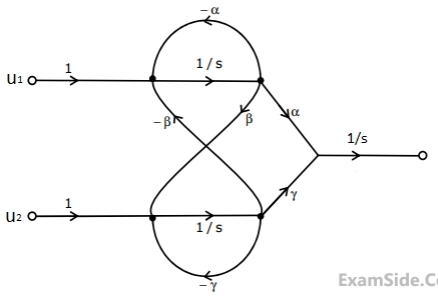
(A) $\frac{s+1}{s^2+1}$ (B) $\frac{s-1}{s^2+1}$ (C) $\frac{s+1}{s^2+s+1}$ (D) $\frac{s-1}{s^2+s+1}$

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Q26. GATE ECE 2008

Open solution

A signal flow graph of a system is given below.



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The set of equations that correspond to this signal flow graph is

(A) $\frac{d}{dt}\begin{bmatrix}x_1\\x_2\\x_3\end{bmatrix} = \begin{bmatrix}\beta & -\gamma & 0\\ \gamma & \alpha & 0\\ -\alpha & -\beta & 0\end{bmatrix}\begin{bmatrix}x_1\\x_2\\x_3\end{bmatrix} + \begin{bmatrix}1 & 0\\0 & 0\\0 & 1\end{bmatrix}\begin{pmatrix}u_1\\u_2\end{pmatrix}$

(B) $\frac{d}{dt}\begin{bmatrix}x_1\\x_2\\x_3\end{bmatrix} = \begin{bmatrix}0 & \alpha & \gamma\\ 0 & -\alpha & -\gamma\\ 0 & \beta & -\beta\end{bmatrix}\begin{bmatrix}x_1\\x_2\\x_3\end{bmatrix} + \begin{bmatrix}0 & 0\\0 & 1\\1 & 0\end{bmatrix}\begin{pmatrix}u_1\\u_2\end{pmatrix}$

(C) $\frac{d}{dt}\begin{bmatrix}x_1\\x_2\\x_3\end{bmatrix} = \begin{bmatrix}-\alpha & \beta & 0\\ -\beta & -\gamma & 0\\ \alpha & \gamma & 0\end{bmatrix}\begin{bmatrix}x_1\\x_2\\x_3\end{bmatrix} + \begin{bmatrix}1 & 0\\0 & 1\\0 & 0\end{bmatrix}\begin{pmatrix}u_1\\u_2\end{pmatrix}$

(D) $\frac{d}{dt}\begin{bmatrix}x_1\\x_2\\x_3\end{bmatrix} = \begin{bmatrix}-\gamma & 0 & \beta\\ \gamma & 0 & \alpha\\ -\beta & 0 & -\alpha\end{bmatrix}\begin{bmatrix}x_1\\x_2\\x_3\end{bmatrix} + \begin{bmatrix}0 & 1\\0 & 0\\1 & 0\end{bmatrix}\begin{pmatrix}u_1\\u_2\end{pmatrix}$

Q27. GATE ECE 2007

Open solution

Consider a linear system whose state space Representation is $\dot{x}(t) = AX(t)$.

If the initial state vector of the system is $x(0) = \begin{bmatrix}1\\-2\end{bmatrix}$,

then the system response is $x(t) = \begin{bmatrix}e^{-2t}\\-2e^{-2t}\end{bmatrix}$.

If the initial state vector of the system changes to $x(0) = \begin{bmatrix}1\\-1\end{bmatrix}$,

then the system response becomes $x(t) = \begin{bmatrix}e^{-t}\\-e^{-t}\end{bmatrix}$.

The eigen value and eigen vector pairs (λ_i, V_i) for the system are

(A) $\left[-1, \begin{bmatrix}1\\-1\end{bmatrix}\right] and \left[-2, \begin{bmatrix}1\\-2\end{bmatrix}\right]$

(B) $\left[-2, \begin{bmatrix}1\\-1\end{bmatrix}\right] and \left[-1, \begin{bmatrix}1\\-2\end{bmatrix}\right]$

(C) $\left[-1, \begin{bmatrix}1\\-1\end{bmatrix}\right] and \left[2, \begin{bmatrix}1\\-2\end{bmatrix}\right]$

(D) $\left[1, \begin{bmatrix}1\\-1\end{bmatrix}\right] and \left[-2, \begin{bmatrix}1\\-2\end{bmatrix}\right]$

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Q28. GATE ECE 2007

Open solution

Consider a linear system whose state space Representation is $\dot{x}(t) = AX(t)$.

If the initial state vector of the system is $x(0) = \begin{bmatrix} 1 \\ -2 \end{bmatrix}$,

then the system response is $x(t) = \begin{bmatrix} e^{-2t} \\ -2e^{-2t} \end{bmatrix}$.

If the initial state vector of the system changes to $x(0) = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$,

then the system response becomes $x(t) = \begin{bmatrix} e^{-t} \\ -e^{-t} \end{bmatrix}$.

The system matrix a is

(A) $\begin{bmatrix} 0 & 1 \\ -1 & 1 \end{bmatrix}$

(B) $\begin{bmatrix} 1 & 1 \\ -1 & -2 \end{bmatrix}$

(C) $\begin{bmatrix} 2 & 1 \\ -1 & -1 \end{bmatrix}$

(D) $\begin{bmatrix} 0 & 1 \\ -2 & -3 \end{bmatrix}$

Q29. GATE ECE 2007

Open solution

The state space representation of a separately excited **DC** servo motor dynamics is given as

$$\begin{bmatrix} \frac{d\omega}{dt} \\ \frac{di_a}{dt} \end{bmatrix} = \begin{bmatrix} -1 & 1 \\ -1 & -10 \end{bmatrix} \begin{bmatrix} \omega \\ i_a \end{bmatrix} + \begin{bmatrix} 0 \\ 10 \end{bmatrix} u.$$

Where 'ω' is the speed of the motor, 'i_a' is the armature current and u is the armature voltage. The transfer function $\frac{\omega(s)}{U(s)}$ of the motor is

(A) $\frac{10}{s^2+11s+11}$

(B) $\frac{1}{s^2+11s+11}$

(C) $\frac{10s+10}{s^2+11s+11}$

(D) $\frac{1}{s^2+s+11}$

Q30. GATE ECE 2006

Open solution

A linear system is described by the following state equation

$$\dot{x}(t) = AX(t) + BU(t), A = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}.$$

The state-transition matrix of the system is

(A) $\begin{bmatrix} \cos t & \sin t \\ -\sin t & \cos t \end{bmatrix}$

(B) $\begin{bmatrix} -\cos t & \sin t \\ -\sin t & -\cos t \end{bmatrix}$

(C) $\begin{bmatrix} -\cos t & -\sin t \\ -\sin t & \cos t \end{bmatrix}$

(D) $\begin{bmatrix} \cos t & -\sin t \\ \cos t & \sin t \end{bmatrix}$

Q31. GATE ECE 2004

Open solution

The state variable equations of a system are: $\dot{x}_1 = -3x_1 - x_2 + u\ \dot{x}_2 = 2x_1\ y = x_1 + u.$

The system is

(A) controllable but not observable.

(B) observable but not controllable.

(C) neither controllable nor observable.

(D) controllable and observable.

Q32. GATE ECE 2004

Open solution

Given A = $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$, the state transition matrix e^{At} is given by

(A) $\begin{bmatrix} 0 & e^{-t} \\ 0^{-t} & 0 \end{bmatrix}$

(B) $\begin{bmatrix} e^{t} & 0 \\ 0 & e^{t} \end{bmatrix}$

(C) $\begin{bmatrix} e^{-t} & 0 \\ 0 & e^{-t} \end{bmatrix}$

(D) $\begin{bmatrix} 0 & e^{t} \\ e^{t} & 0 \end{bmatrix}$

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Q33. GATE ECE 2004

If A = $\begin{bmatrix} -2 & 2 \\ 1 & -3 \end{bmatrix}$, then sin At is

(A) $\frac{1}{3}\begin{bmatrix} \sin(-4t)+2\sin(-t) & -2\sin(-4t)+2\sin(-t) \\ -\sin(-4t)+\sin(-t) & 2\sin(-4t)+\sin(-t) \end{bmatrix}$

(B) $\begin{bmatrix} \sin(-2t) & \sin(2t) \\ \sin(t) & \sin(-3t) \end{bmatrix}$

(C) $\frac{1}{3}\begin{bmatrix} \sin(4t)+2\sin(t) & 2\sin(-4t)-2\sin(-t) \\ -\sin(-4t)+\sin(t) & 2\sin(4t)+\sin(t) \end{bmatrix}$

(D) $\frac{1}{3}\begin{bmatrix} \cos(-t)+2\cos(t) & 2\cos(-4t)+2\sin(-t) \\ -\cos(-4t)+\sin(-t) & -2\cos(-4t)+\cos(-t) \end{bmatrix}$

Open solution

Q34. GATE ECE 2003

The zero, input response of a system given by the state space equation

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} and \begin{bmatrix} x_1 & (0) \\ x_2 & (0) \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \end{bmatrix} is$$

(A) $\begin{bmatrix} te^t \\ t \end{bmatrix}$

(B) $\begin{bmatrix} e^t \\ t \end{bmatrix}$

(C) $\begin{bmatrix} e^t \\ te^t \end{bmatrix}$

(D) $\begin{bmatrix} t \\ te^t \end{bmatrix}$

Open solution

Q35. GATE ECE 1999

For the system described by the state equation

$$\dot{x} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0.5 & 1 & 2 \end{bmatrix}x + \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}u.$$

If the control signal u is given by u=(-0.5-3-5)x+v, then the eigen values of the closed loop system will be

(A) 0, -1, -2 (B) 0, -1, -3 (C) -1, -1, -2 (D) 0, -1, -1

Open solution

Q36. GATE ECE 1997

A certain linear time invariant system has the state and the output equations given below

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} 1 & -1 \\ 0 & 1 \end{bmatrix}\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} 0 \\ 1 \end{bmatrix}u$$

$$y = \begin{bmatrix} 1 & 1 \end{bmatrix}\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}, if$$

$x_1(0) = 1, x_2(0) = -1, u(0) = 0$, then $\frac{dy}{dt}|_{t=0}$ is

(A) 1

(B) -1

(C) 0

(D) None of the above

Open solution

Q37. GATE ECE 1992

A linear time-invariant system is described by the state variable model

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} -1 & 0 \\ 0 & -2 \end{bmatrix}\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} 0 \\ 1 \end{bmatrix}u.$$

$$Y = \begin{bmatrix} 1 & 2 \end{bmatrix}\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

(A) The system is completely controllable

(B) The system is not completely controllable

(C) The system is completely observable

(D) The system is not completely observable

Open solution

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Q38. GATE ECE 1991

Open solution

A linear second order single input continuous-time system is described by the following set of differential equations

$$\dot{x_1}(t) = -2x_1(t) + 4x_2(t)$$
$$\dot{x_1}(t) = 2x_1(t) - x_2(t) + u(t)$$

Where $x_1(t)$ and $x_2(t)$ are the state variables and u (t) is the control variable. The system is

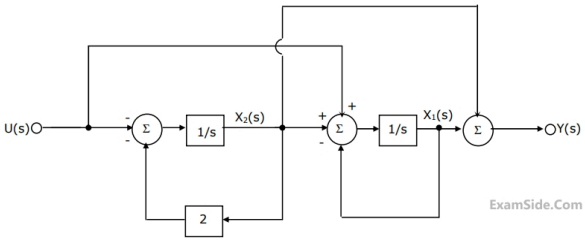
(A) controllable and stable.
(B) controllable but unstable.
(C) uncontrollable and unstable.
(D) uncontrollable but stable.

Marks 5 4 questions

Q39. GATE ECE 2002

Open solution

The block diagram of a linear time invariant system is given in Figure is



(a) Write down the state variable equations for the system in matrix form assuming the state vector to be $[x_1(t)\ x_2(t)]^T$
(b) Find out the state transition matrix.
(c) Determine y(t), t ≥ 0, when the initial values of the state at time t = 0 are $x_1(0) = 1$, and $x_2(0) = 1$.

Q40. GATE ECE 2000

Open solution

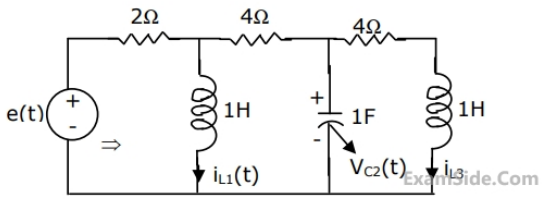
A certain linear, time-invariant system has the state and output representation shown below:

$$\begin{bmatrix}\dot{x_1}\\ \dot{x_2}\end{bmatrix} = \begin{bmatrix}-2 & 1\\ 0 & -3\end{bmatrix}\begin{bmatrix}x_1\\ x_2\end{bmatrix} + \begin{bmatrix}1\\ 0\end{bmatrix}u$$
$$y = \begin{pmatrix}1 & 1\end{pmatrix}\begin{bmatrix}x_1\\ x_2\end{bmatrix}$$

(a) Find the eigen values (natural frequencies) of the system.
(b)If u(t)=$\delta(t)$ and $x_1(0^+)=x_2(0^+)=0$, find $x_1(t)$,$x_2(t)$ and y(t), for t>0.
(c)When the input is zero, choose initial conditions $x_1(0^+)$ and $x_2(0^+)$ such that $y(t) = Ae^{-2t}$ for t>0

Q41. GATE ECE 1997

Open solution



For the circuit shown in the figure, choose state variables as x_1, x_2, x_3 to be $i_{L1}(t), v_{c2}(t), i_{L3}(t)$
Wriote the state equations

$$\begin{bmatrix}\dot{x_1}\\ \dot{x_2}\\ \dot{x_3}\end{bmatrix} = A\begin{bmatrix}x_1\\ x_2\\ x_3\end{bmatrix} + B[e(t)]$$

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Q42. GATE ECE 1996

Obtain a state space representation in diagonal form for the following system

$$\frac{d^3y}{dt^3} + 6\frac{d^2y}{dt^2} + 11\frac{dy}{dt} + 6y = 6u(t)$$

Open solution

Marks 10 1 questions

Q43. GATE ECE 1990

A system is characterized by the following state-space equations:

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} -3 & 1 \\ -2 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} 0 \\ 1 \end{bmatrix} u; t > 0$$

$$y = \begin{bmatrix} 1 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

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Open solution

a) Find the transfer function of the system

b) Compute the state-transition matrix/p>

a) Solve the state equation for a unit-step input under zero initial conditions

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Answer Key

Marks 1

1. **5**	2. **C**	3. **B**	4. **D**	5. **A**

Marks 2

6. **C**	7. **B**	8. **A, C**	9. **A**	10. **A**
11. **D**	12. **A**	13. **A**	14. **D**	15. **B**
16. **D**	17. **C**	18. **A**	19. **B**	20. **A**
21. **A**	22. **D**	23. **B**	24. **B**	25. **C**
26. **D**	27. **A**	28. **D**	29. **A**	30. **A**
31. **D**	32. **B**	33. **A**	34. **C**	35. **A**
36. **A**	37. **B, C**	38. **B**		

Marks 5

39.

$$a)\begin{bmatrix}\dot{x}_1(t)\\ \dot{x}_2(t)\end{bmatrix}=\begin{bmatrix}1 & e^{-2t}\\ 0 & -2\end{bmatrix}\begin{bmatrix}x_1(t)\\ x_2(t)\end{bmatrix}+\begin{bmatrix}1\\ -1\end{bmatrix}\mu(t)$$

$$y(t)=\begin{bmatrix}1 & 1\end{bmatrix}\begin{bmatrix}x_1(t)\\ x_2(t)\end{bmatrix}$$

$$b)\phi(t)=\begin{bmatrix}e^{-2t} & e^{-t}-e^{-2t}\\ 0 & e^{-2t}\end{bmatrix}$$

$$c)\,y(t)=e^{-t}+e^{-2t}$$

40. **(a)Eiger values are=-2, -3**
(b)
$x_1(t)=1$
$x_2(t)=0$
$y(t)=e^{-2t}$
(c) $x_1(0^+)=A$ **and** $x_2(0^+)=0$

41. **Solve it.**

42.

$$\begin{bmatrix}\dot{x}1\\ \dot{x}_2\\ \dot{x}_3\end{bmatrix}=\begin{bmatrix}-1 & 0 & 0\\ 0 & -2 & 0\\ 0 & 0 & -3\end{bmatrix}\begin{bmatrix}x_1\\ x_2\\ x_3\end{bmatrix}+\begin{bmatrix}t\\ t\\ t\end{bmatrix}$$

$$y(s)=\begin{bmatrix}1 & 1 & 1\end{bmatrix}\begin{bmatrix}x_1\\ x_2\\ x_3\end{bmatrix}$$

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Marks 10

43. **a)** $T.F=\frac{1}{(s+1)(s+2)}$ **b) State transition matrix**

$$\begin{bmatrix}-e^{-t}+2e^{-2t} & e^{-t}-e^{-2t}\\ -2e^{-t}+2e^{-2t} & 2e^{-t}-e^{-2t}\end{bmatrix}$$

c) State Equation

$$\begin{bmatrix}\frac{1}{2}u(t)-e^{-t}+\frac{1}{2}e^{-2t}\\ \frac{3}{2}u(t)-2e^{-t}+\frac{1}{2}e^{-2t}\end{bmatrix}$$

For numerical answers given as a range, any value within the range is correct. Full solutions: scan the QR code next to each question, or visit questions.examside.com.

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